

① IP: 192.168.10.0

Subnet Mask: 255.255.255.224

② Block size?

We know, Block size = $256 - \text{Last Octet of Mask}$

$$= 256 - 224$$

$$= 32$$

This means subnets increase by 32 in the last octet:

0, 32, 64, 96, 128, 160, 192, 224

③ Number of Subnets?

Given,

Subnet Mask = 255.255.255.224

Using a calculator,

$$(255)_{10} = (11111111)_2$$

$$(224)_{10} = (11100000)_2$$

∴ Converting the Subnet Mask to binary:

11111111. 11111111. 11111111. 11100000

We have 3 1's in the host octet.

$$\therefore \text{Number of Subnets} = 2^3 = 8$$

④ Number of Hosts?

$$\begin{aligned} \text{Number of Hosts} &= 2^y \quad [y = \text{number of 0's in host octet}] \\ &= 2^5 = 32 \end{aligned}$$

$$\therefore \text{Valid hosts} = 32 - 2 = 30$$

④ What is the network address?

Given IP address = 192.168.10.0

Block size = 32

∴ Block ranges :

0 → 0-31

32 → 32-63

64 → 64-95

96 → 96-127

128 → 128-159

160 → 160-191

192 → 192-223

224 → 224-255

192.168.10.0 falls in the range 0-31

Therefore,

Network address = 192.168.10.0

⑤ What is the Broadcast address?

Broadcast = Last number of subnet range

Since 192.168.10.0 falls under the range 0-31

∴ Broadcast address = 192.168.10.31

⑥ Get first and last valid host addresses.

First host = Network + 1

= 192.168.10.1

Last host = Broadcast - 1

= 192.168.10.30

- ② Your organization has been assigned a class A IP Address of 10.0.0.0

① Total Host Addresses

Since the IP address of the organization is 10.0.0.0 which is a class A IP address,

Subnet mask : 255.0.0.0 (Default)

Prefix length : /8

Host bits : $32 - 8 = 24$

So, there are 2^{24} host addresses, where $2^{24} - 2$ of them are valid.

② Total Subnet Addresses

For a default class A network (no subnetting applied), there is only 1 subnet address which is 255.0.0.0

③ First address & First valid host address

First network address : 10.0.0.0

First valid host : 10.0.0.1

④ Last address & Last valid host address

Last network address : 10.255.255.255

Last valid host : 10.255.255.254

© Addresses of 55,10,280th, 85,259th & 6,573th

From the given scenario, the first 8 bits of the 32-bit address is fixed (10).

Binary of 10 = 00001010

∴ Network address = 10.A.B.C

Here, A-C denotes the remaining 24 bits of the address. Since the first valid host starts at 1, we need to subtract 1 from the Nth host.

55, 10, 280th

We have 5510280

$$\Rightarrow 5510280 - 1 = 5510279$$

Dividing by 2^{16} :

$$5510279 \div 2^{16} = 84 \text{ remainder } 5255$$

$$\therefore A = 84, \quad B = 5255$$

Dividing by 2^8 :

$$5255 \div 2^8 = 20 \text{ remainder } 135$$

$$\therefore B = 20, \quad C = 135$$

$$\therefore \text{IP address} = \underset{A}{10} . \underset{B}{84} . \underset{C}{20} . 135$$

	2^{16}	2^8	2^0
10	A	B	C

85,259th

We have 85259

$$\Rightarrow 85259 - 1 = 85258$$

Dividing by 2^{16} :

$$85258 \div 2^{16} = 1 \text{ remainder } 19722$$

$$\therefore A = 1, r = 19722$$

Dividing by 2^8 :

$$19722 \div 2^8 = 77 \text{ remainder } 10$$

$$\therefore B = 77, C = 10$$

$$\therefore \text{IP address} = 10 \cdot 1 \cdot 77 \cdot 10$$

A B C

6,573th

We have 6573

$$\Rightarrow 6573 - 1 = 6572$$

Dividing by 2^{16} :

$$6572 \div 2^{16} = 0 \text{ remainder } 6572$$

$$\therefore A = 0, r = 6572$$

Dividing by 2^8 :

$$6572 \div 2^8 = 25 \text{ remainder } 172$$

$$\therefore B = 25, C = 172$$

$$\therefore \text{IP Address} = 10 \cdot 0 \cdot 25 \cdot 172$$

A B C